

Hepatic Disorders of Sauropsids

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INTRODUCTION

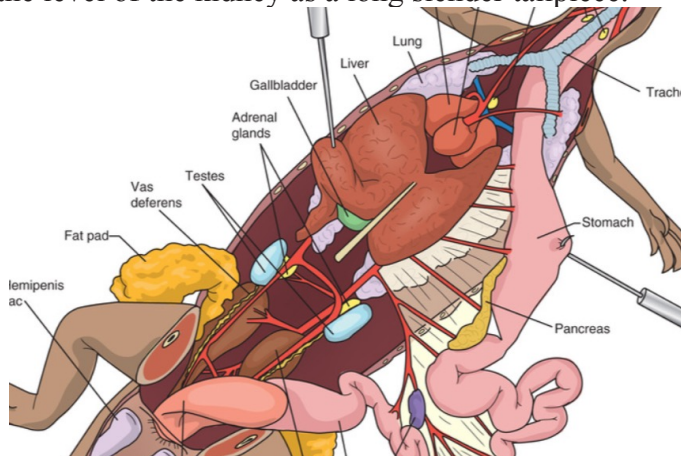
As in mammals, the liver plays a central role in the physiology and metabolism of both birds and reptiles. As such, it is **frequently affected by systemic illnesses, particularly multisystemic infectious diseases**. Clinical signs associated with hepatic dysfunction are often varied and non-specific, commonly including anorexia, lethargy, and weight loss, among others.

In this lecture, we will begin with a review of key background concepts you have encountered in previous courses, now reframed through a liver-focused lens to provide context for understanding hepatic diseases and their clinical workup.

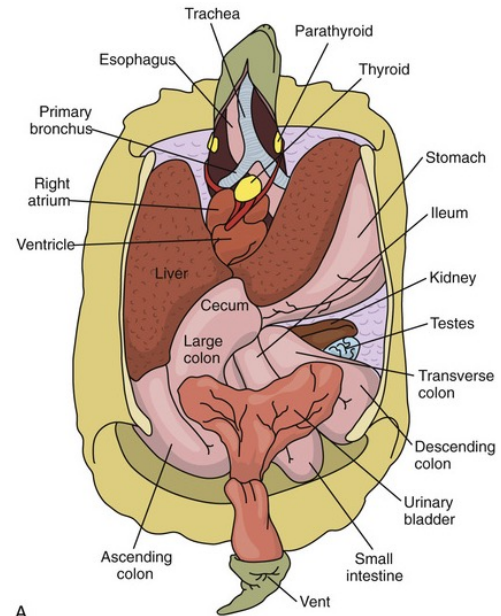
ANATOMY - RECAP

Reptiles

The reptilian liver is composed of **2 lobes in lizards and chelonians with the right generally being bigger and harboring the gallbladder** and surrounding the heart. In some species, notably our beloved bearded dragons, the right liver lobe extends very caudally to the level of the kidney as a long slender tailpiece.

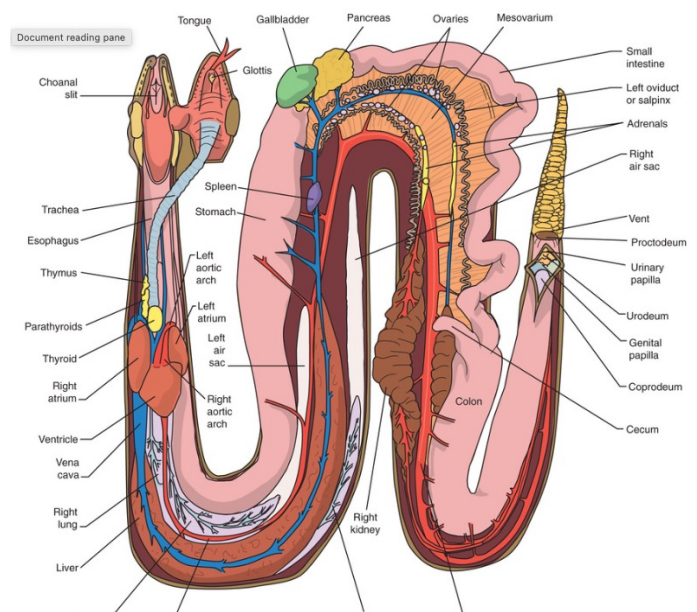


In chelonians, the liver is transverse and occupy the majority of the cranial coelomic cavity.

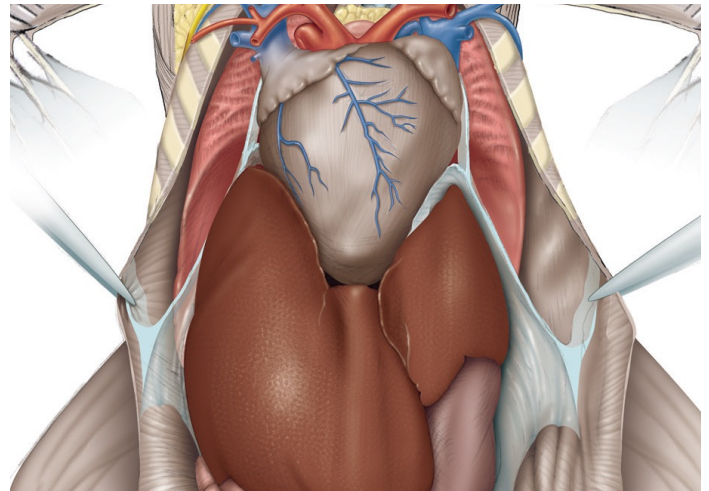


A

In snakes, the liver is unlobed and the gallbladder is located caudally to the liver.



The coelomic location of the liver varies in reptiles depending on the species as coelomic cavity organization varies in reptiles. In most squamates, the coelomic cavity is undivided in a single pleuroperitoneal cavity, therefore the liver is just floating in there. In some lizards, there are additional separations. In monitor lizards, there is a **postpulmonary septum**, therefore the liver is in the peritoneal cavity (the other cavity being the pleural cavity). In tegus, there is a **posthepatic septum** splitting the cavity in two, a thoracic cavity where the liver is (I know it's weird), and an abdominal cavity. In many chelonians, a **postpulmonary septum** is also present and the liver is consequently in the peritoneal cavity. Crocodilians are a bit different and more like birds (they are archosaurs), therefore the liver is in the hepatoperitoneal cavities.

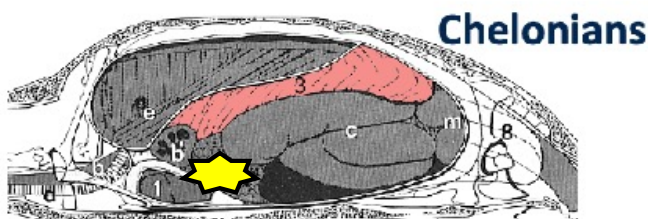
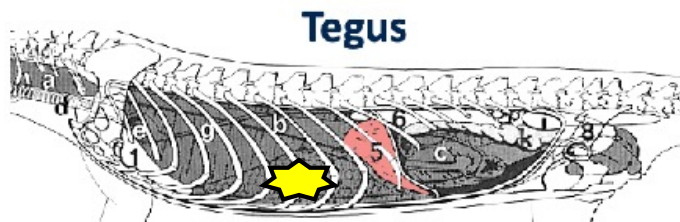


The liver is surrounding the heart, like in reptiles, and is located within the **4 hepatoperitoneal cavities**. These cavities are midline right below the keel. They are bordered laterally by the **oblique septum** and caudally by the **posthepatic septum**. They are split in the middle by the **ventral mesentery**.

HISTORY AND CLINICAL SIGNS

It is first important to inquire about the duration and severity of clinical signs and review husbandry for evidence to risk factors exposure. Risk factors for infectious diseases may include a non-quarantined animal, a recently purchased animal, or exposure to an infected animal or a species known to carry a particular virus. Risk factors for non-infectious diseases depend on the condition but obesity, **high-fat nutritionally deficient diets**, **high-iron diet for susceptible species**, **inactivity**, and **low temperature in reptiles** may all predispose to specific metabolic or developmental disorders.

History and clinical signs are primarily characterized by **non-specific signs** such as anorexia and lethargy. Polyuria, diarrhea and soft stools can also be encountered. **Biliverdinuria**, manifesting by green urine in the urofeces, may occur in moderate to severe cases.



Birds

Similarly to reptiles, birds generally have **two liver lobes**, with the right being bigger and harboring the **gallbladder**, when present. There are many species in which the gallbladder is absent. **In psittacine birds, the gallbladder is absent in parrots, but present in cockatoos (Cacatuidae family)**. The gallbladder is generally present in carnivorous birds such as most raptor species. It is absent in pigeons and ostriches, but present in emus. It is surprisingly long in toucans.



As a reminder, birds and reptiles **lack significant biliverdin reductase** so they only produce low quantity of bilirubin. Therefore, **we don't see jaundice or icterus in birds and reptiles** and we rather see biliverdinuria. The yellow coloration of the plasma is attributed to carotenoids. Bearded dragons also naturally have yellow oral mucus membranes, which should not be confused with icterus.

In most lizards, the liver is in the same cavity than all other organs, so advanced hepatomegaly may lead to signs that would not be expected in other species such as dyspnea from pressing on the lungs and regurgitation from pressing on the GI system.

With severe hepatic dysfunction, signs of hepatic insufficiency may be seen such as **coelomic effusion, metabolic disruptions, and coagulation disorders**. Hepatic encephalopathy seems to be extremely rare overall in birds and reptiles. While it has not been conclusively demonstrated in birds, there are a few convincing cases in reptiles.

DIAGNOSTIC WORKUP

Like for any significant presenting condition, a **minimum diagnostic database** of blood work (CBC, biochemistry) and survey imaging (radiographs, CT-scan) should be obtained. Infectious disease differentials should be explored using disease-specific testings such as serology and molecular testing. Bacterial and fungal cultures may also be helpful. Blood culture may be attempted as well as bacterial hepatitis tend to also be associated with some degree of bacteremia.

Ultimately, a confirmed diagnosis has to be obtained through **liver sampling**. Fine-needle aspiration and cytology from an ultrasound-guided sampling approach is the least invasive way of doing this. **Liver biopsy through laparotomy or coelioscopy** is also an option, but is more invasive.

Biochemistry

The biochemistry panel and interpretation are somewhat similar between birds and reptiles with a few species-specific features. Please also consult your **clinical pathology lecture and lab** for more information on

this topic. One should always keep in mind that **tissue enzyme activities have only been determined in a handful of species** and while a common pattern seems to prevail, there are many species that have specific tissue distribution that would change the biochemistry interpretation. Also, half-lives in the plasma are important to consider when known. It is for instance extremely short for CK and a bit longer for AST in most species. **Since there are no enzyme that is both sensitive and specific and each give slightly different information, a combination of enzymes and other liver parameters is necessary to assess liver integrity**. As a rule, AST and LDH are sensitive but not specific, so their elevation needs to coincide with a normalish CK (specific to muscle) to be interpreted as hepatocellular leakage. **GLDH is not sensitive, but is an extremely specific enzyme to the liver**. It gets elevated in severe hepatic lesions such as hepatic necrosis or advanced hepatitis. GGT is an induced enzyme that tends to get elevated with cholestasis, but its usefulness is debated.

Previous intramuscular injections may result in a plasma increase of AST, LDH, and CK lasting several days, further complicating the interpretation of an already complex liver profile. So think about it before sedating with IM drugs.

Liver enzymes overall tend to have a **low sensitivity for chronic metabolic disorders** such as hepatic lipodosis, iron storage disease, and amyloidosis.

Simplified Tissue Distribution of Commonly Used Enzymes in Avian Biochemistry ^{19,38,39}							
	Hepatic Tissue			Muscular Tissue			Other Tissues
	Detection	Sensitivity	Specificity	Detection	Sensitivity	Specificity	Detection
AST	++	+++	-	++	+++	-	-
GLDH	+++	+	+++	+	-	-	+++ (kidney but excreted in urine)
GGT	+	+	+++	-	-	-	+++ (kidney)
LDH	++	+	-	++	+	-	+
ALT	++	++	-	++	+++	-	+
ALP	+	-	-	++	-	-	+
CK	-	-	-	+++	+++	+++	-
NAG	+	-	-	-	-	-	+++ (6 times higher in kidneys)

Bile acids are an essential analyte to include in hepatic biochemistry panel in both birds and reptiles. It reflects hepatic function (so not leakage) and is considered to be **both sensitive and specific**, so probably the best hepatic biomarker on your panel. Post-prandial increase occurs in both birds and reptiles regardless of their diet. **The upper limit of the reference interval tends to be much lower in reptiles** (around 20-30 mmol/L compared to 70 in birds), so make sure you use appropriate references.

As bilirubin is minimally present in both birds and reptiles, we typically do not include a bilirubin test on

the panel. Measuring biliverdin would be great, but it is a very labile analyte and excreted quickly through the kidney, so it is not currently deemed measurable or useful. Again, remember that yellow plasma in birds and reptiles is due to other pigments, mainly carotenoids.

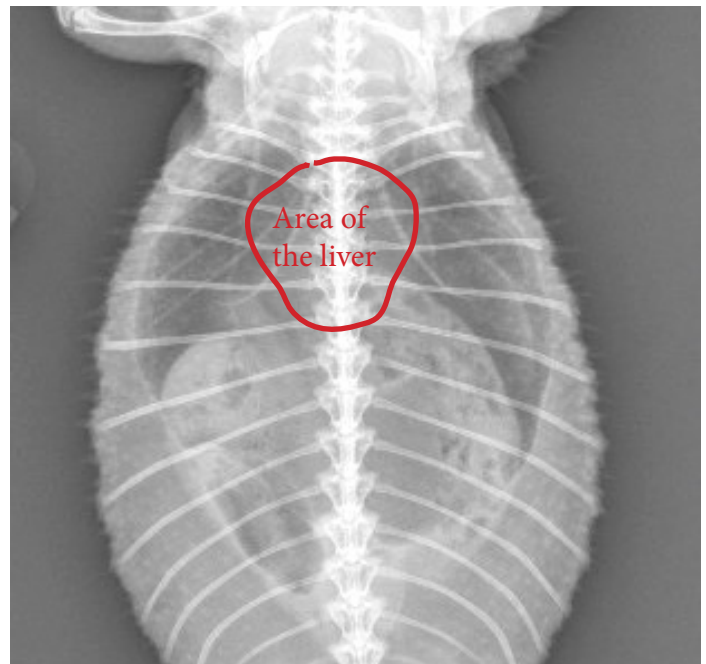
Ammonia is typically not measured as it is highly volatile and hepatic encephalopathy is considered a rare occurrence.

Metabolites present on the biochemistry panel may also give valuable information regarding hepatic function. Uric acid, glucose, and proteins are made in the liver and will decrease with hepatic insufficiency, but typically more when it is end-stage, so not commonly. Cholesterol is also made in the liver, but is excreted through the bile, so **biliary stasis will lead to hypercholesterolemia**. High cholesterol may also reflect a general hyperlipidemia associated with other lipid disorders such as hepatic lipidosis (although it does not seem to be the case in most birds and reptiles). β -hydroxybutyric acid, the main ketone produced by birds and reptiles, has been shown to decrease with hepatic lipidosis in bearded dragons.

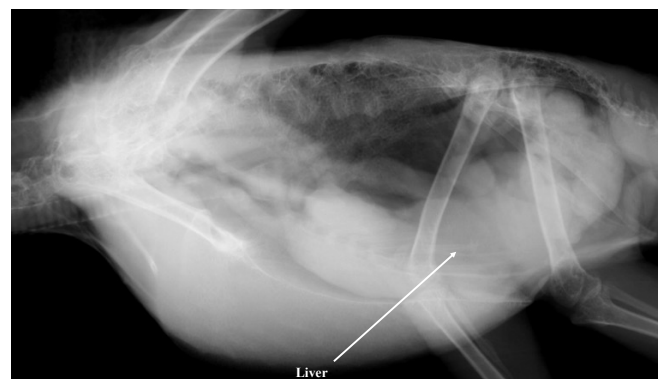
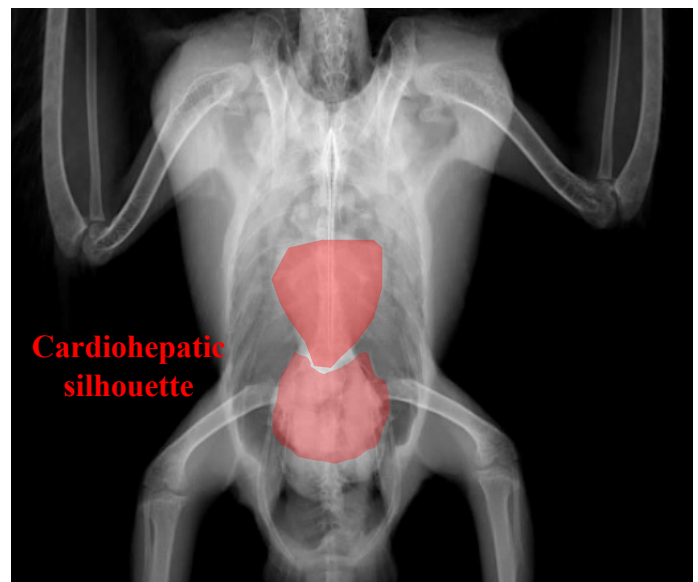
Coagulation testing may also be performed if necessary.

Radiographs

Survey radiographs are not super helpful in reptiles overall owing to the poor soft tissue contrast in the coelomic cavity. So only the region of the liver can be seen, but hepatomegaly itself may not be easily interpretable.



In birds, there is a bit more contrast owing to the air sac system. The liver is mainly assessed in the ventro-dorsal radiographic view. It blends with the heart cranially in an hourglass shaped shadow that people refer to as the **cardiohepatic silhouette**. The caudal part of that hourglass should **not extend laterally too much passed the acetabula**.

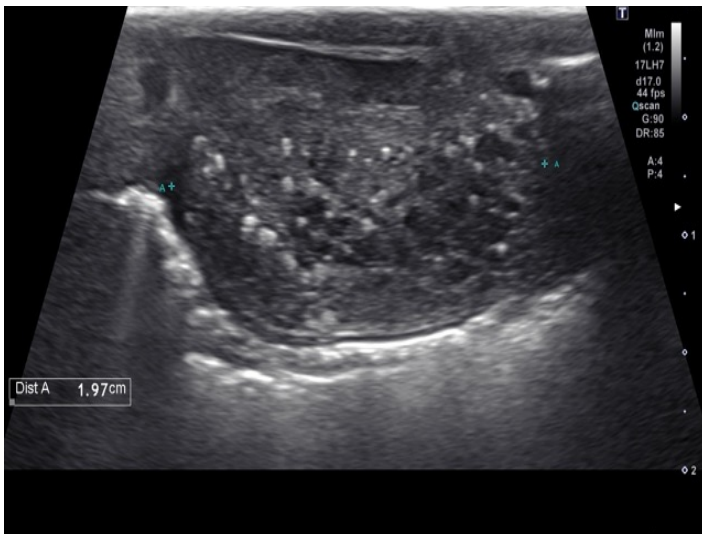


On the lateral view, an enlarged liver will push adjacent structures dorsally, typically the GI and spleen. The spleen is easily visible on the lateral view and with infectious hepatitis, it is also frequently enlarged (splenohepatitis).

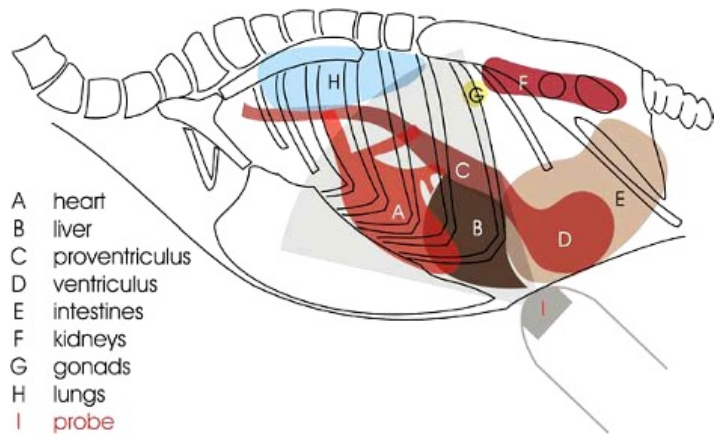
Ultrasound

Ultrasound of the coelom is super useful in reptiles. I actually recommend **performing a point-of-care coelomic ultrasound in most reptiles as part of the physical examination**. The liver is easily visualized in most species midline and also in large chelonians through the prefemoral fossa. **Echostructure, echogenicity, the presence of masses and their vascular pattern are all used to assess the liver for lesions**. Echogenicity is overall a bit difficult to assess in reptiles because of a lack of reference organ, but the fat pads can be used as a reference for an hyperechoic organ. The nice thing with ultrasound is **the ability to obtain targeted fine-needle aspiration for cytology**. It is relatively cheap and often rewarding at diagnosing most diseases.

Below is an ultrasound of a bearded dragon with a liver mass.



In birds, ultrasound is a bit **more limited due to their extensive air sac system, but also due to the large keel** that shields most of the cranial coelomic organ including the liver. **When placing a small probe midline right passed the keel and angling cranially, the liver is typically visible**. It is one of the few organs that can reliably be imaged by ultrasound in birds. Coelomic effusion and hepatomegaly make hepatic ultrasound investigations easier.



CT-Scan

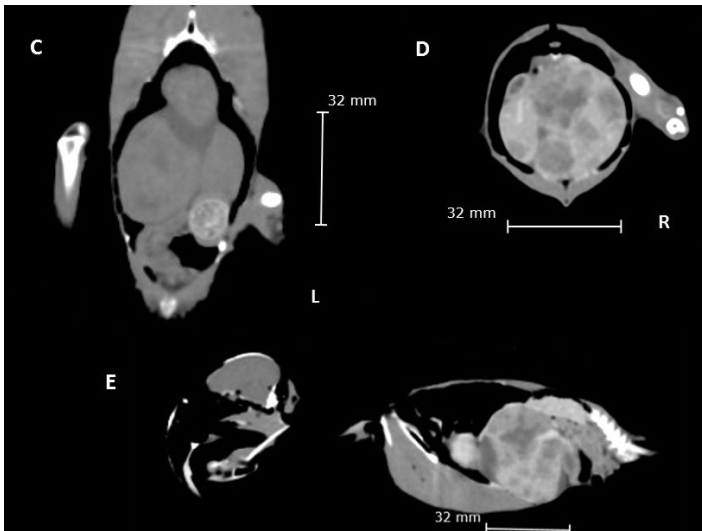
CT-Scan is great in both reptiles and birds. Reptiles have poor soft tissue contrast on radiographs and the shell also decreased overall contrast in chelonians. The liver (and other organs) are easier to see on contrast CT-scan. One of the best thing when it comes to the liver is the ability to obtain its **density**, which is useful in the case of fat accumulation (low density with hepatic lipidosis), decrease water presence (high density with fibrosis) and accumulation of high atomic number compounds (iron). Masses can also be easily diagnosed as well as their potential nature depending on whether they are contrast enhancing. The gallbladder is also seen in most species.

Below is cross-section of a tortoise with advanced hepatic fibrosis with nodular regeneration. See the high density of the liver compared to the heart (center) and most of the other organs.



See later for CT-scan of lipidotic liver.

Same for birds, CT is much better than xray generally and the liver is easily visualized. Below is a bird with heterogeneous liver due to neoplasias.

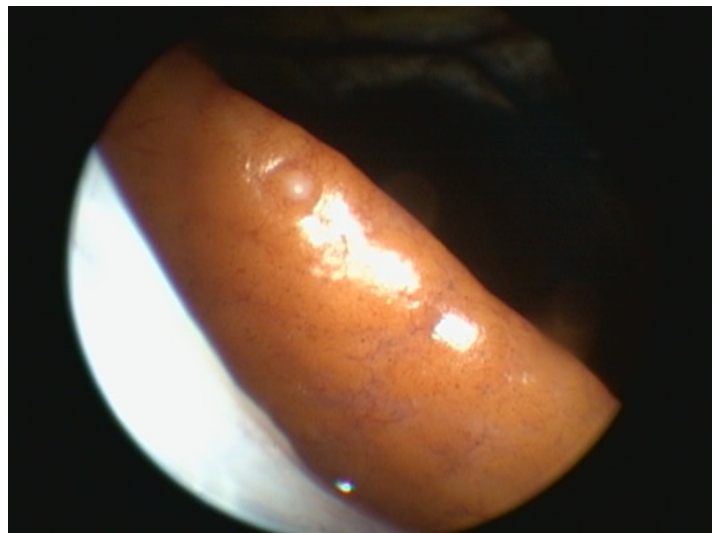


Endoscopy and biopsy

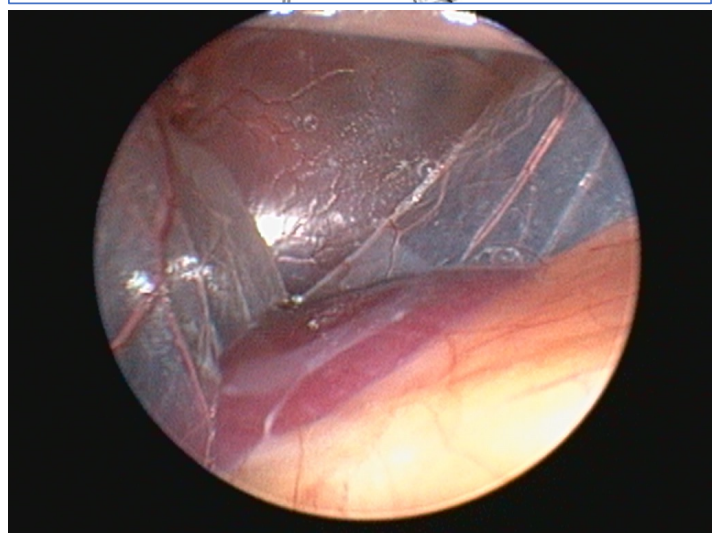
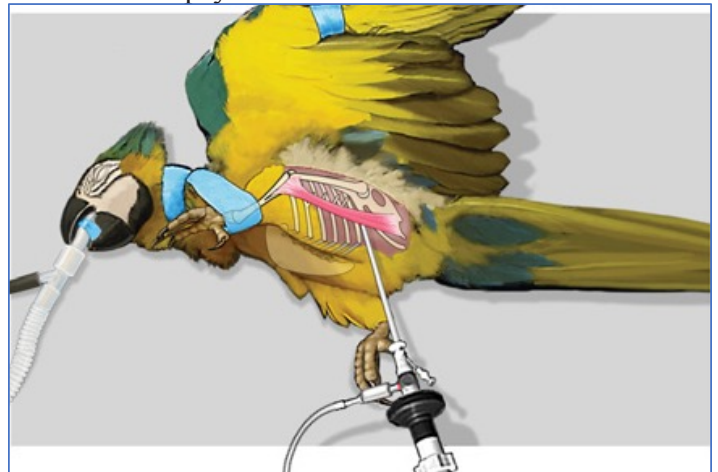
Ultrasound of the coelom is incredibly valuable in both reptiles and birds. The techniques and coelomic approaches are a bit different, but the overall equipment and principles are similar. **A 2.5 mm 30 degree rigid endoscope is used with an operating sheath and small 5Fr cup biopsy forceps.** This allow to obtain targeted biopsies from lesions.

In lizards, you need to insufflate the coelom with CO₂, which can be done through one of the channels of the operating endoscopic sheath. The incision is made paramedian to avoid the central abdominal vein. The liver is typically the first organ that you see and the easiest to sample. Multiple biopsies for histopathology, culture, and PCR are recommended. **For chelonians, the entry point is in the prefemoral fossa.** Sometimes, insufflation is not necessary to just get to the liver as the shell keeps the overall expansion of the coelom.

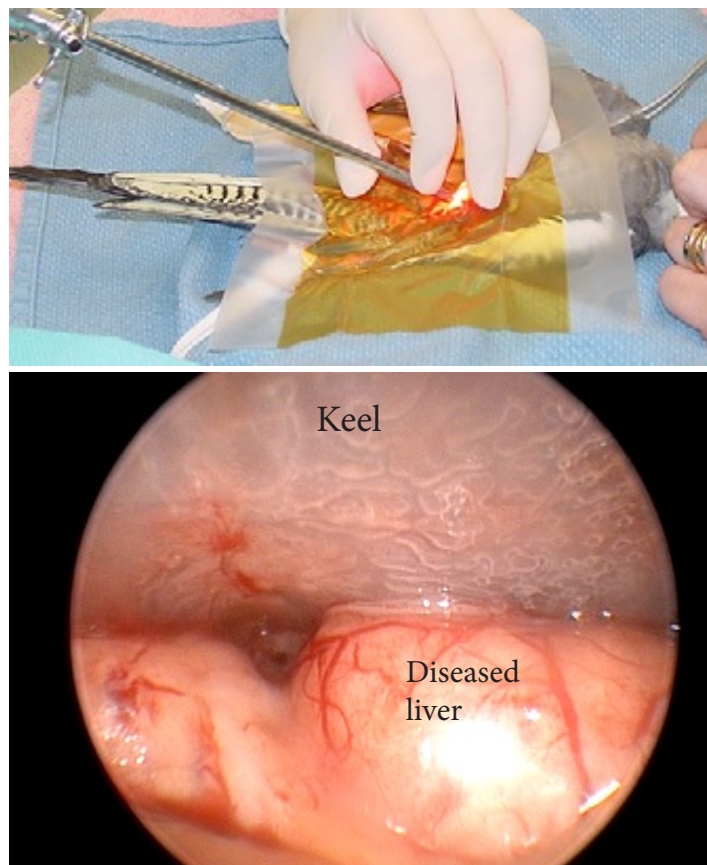
Below is a coelioscopy on a bearded dragon:



In birds, it is a bit more complex because of the extensive air sac system and the complexity of the coelomic cavities. There are essentially **two approaches**. The first one is the **general air sac endoscopy approach through the left or right side**. You see all the other organs including the liver, and you can make a hole in the air sac membrane/oblique septum once in the caudal thoracic or cranial thoracic air sac. From there you can take a biopsy.



The other one is better when just focusing on the liver. It is the midline approach. You make a **tiny incision past the keel**. From there, you may have to perforate the posthepatic septum with mosquito hemostat to end up in the ventral hepatoperitoneal cavities.

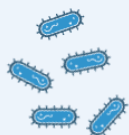


The liver can be sampled from there. It can be hard in small species as there is not a lot of room, so sometimes, it is easier to make the same midline approach and directly take a biopsy of the liver with the forceps under direct visualization (without the scope).

INFECTIOUS HEPATOPATHIES

Because of the liver central role, there are many infectious agents that affect or target the liver as part of a systemic disease/infection. There are few pathogens that specifically target the liver. Below are the general differential diagnosis for infectious hepatopathies in reptiles (left column, shaded with a chameleon) and birds (right column, shaded with a pigeon). In red are the most important or common differentials that I will discuss more in details below. Some diseases were also more extensively covered in other lectures such as avian chlamydiosis in the [respiratory lecture](#) (another great lecture) and chelonians herpesviruses also in the [respiratory lecture](#). Avian mycobacteriosis is also covered in the [GI lecture](#).

Bacteria



- Gram- bacteria
 - *Providencia* spp.
 - *Salmonella* spp.
 - *Citrobacter*
 - *E. coli*
 - *Morganella* spp.
- Reptile chlamydiosis
- Reptile mycobacteriosis
- Gram- bacteria
 - *E. coli*
 - *Klebsiella* spp.
 - *Pseudomonas* spp.
 - *Pasteurella multocida*
 - *Yersinia pseudotuberculosis*
- Gram+ bacteria
 - *Staphylococcus* spp.
 - *Streptococcus* spp.
- Avian chlamydiosis
- Avian coxiellosis
- Avian mycobacteriosis
 - *M. genavense*
 - *M. avium avium*

Parasites



- Intranuclear coccidiosis (INC)
- *Entamoeba invadens*
- *Hepatozoon* spp.
- *Choleoelimeria* spp.
- Systemic coccidiosis
 - *Isospora serini* (atoxoplasmosis)
 - *Eimeria* spp.
- *Histomonas meleagridis*
- *Trichomonas gallinae*
- *Plasmodium* spp.
- *Leucocytozoon simondi*
- *Encephalitozoon hellem*

Fungi



- *Encephalitozoon pogonae*
- *Nannizziopsis* spp.
- *Metarhizium* spp.

Viruses



- Adenoviruses
 - Agamid adenovirus (*Barthadenovirus draconis*)
 - Snake adenovirus (*Barthadenovirus serpentis*)
 - Slider adenovirus (*Testadenovirus trachemysis*)
 - Sulawesi tortoise adenovirus
- Herpesviruses
 - Testudinid herpesviruses (*Scutavirus testudininalpha 3*)
 - Emydid herpesvirus 1
 - Terrapene Herpesvirus 1
- Iridoviruses
 - *Ranavirus rana 1* (Frog virus 3)
- Arenaviruses
 - *Reptarenavirus* spp.
- Adenoviruses
 - Parrot adenovirus (*Aviadenovirus spp.*)
 - Pigeon adenovirus (*Aviadenovirus columbae*)
 - Falcon adenovirus (*Aviadenovirus falconis*)
 - Owl adenovirus (*Aviadenovirus bubonis*)
 - Fowl adenoviruses, inclusion body hepatitis (several)
 - Pheasant marble spleen disease (*Siadenovirus galllopavoterti*)
- Herpesviruses
 - Pacheco's disease, (*Iltovirus psittacidalpha 1*)
 - Duck viral enteritis (*Anatid herpesvirus, Mardivirus anatalpha 1*)
 - Columbid herpesvirus (*Mardivirus columbidalpha 1*)
- Circoviruses
 - PBFD (Circovirus parrot)
- Reoviruses
 - Avian orthoreovirus
- Polyomaviruses
 - Psittacine polyomavirus (*Gammapolyomavirus avis*)
- Hepadnaviruses
 - Duck hepatitis B (*Avihepadnavirus anatiguidae*)
 - Parrot hepatitis B (*Avihepadnavirus psittainius*)
- Picornaviruses
 - Duck hepatitis A (*Avihepatovirus ahepati*)

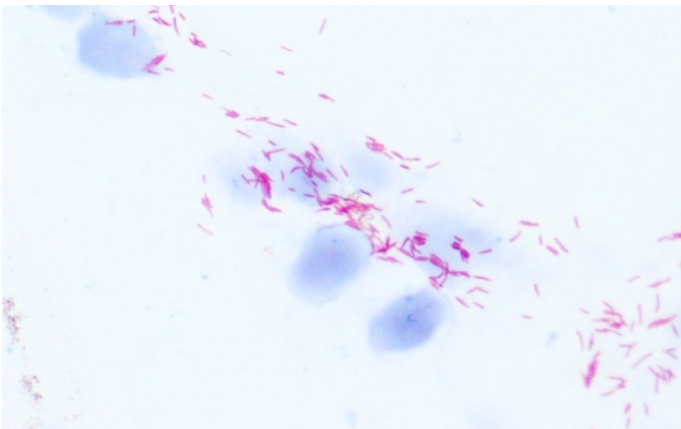
Reptile mycobacteriosis

Reptile mycobacteriosis is a sporadic disease and is not super common, but also not uncommon (well, sporadic...). We are talking with reptiles (and birds) about **non-tuberculous mycobacteria** that cannot cause tuberculosis in people and are mostly saprophytic ubiquitous environmental organisms.

In reptiles, the disease is **most commonly seen in aquatic chelonians** such as red-eared sliders, box turtles and other freshwater turtles. Animals under environmental or immunologic stress seem predisposed.

There is a high number of species that have been implicated with **the most common being *Mycobacterium marinum*. The second most common is *Mycobacterium chelonae***, but many others have been identified (*M. thamnophaeos*, *kansasii*...).

Clinical signs are, you guessed it, non-specific for the most part. The disease is a *chronic granulomatous disease* that can affect many different organs and locations and the liver is just one of those. Nothing specific either on routine diagnostic test other than when obvious lesions are seen on the skin or on the bones or on ultrasound. Lesions have to be sampled for confirmatory diagnosis with either cytology or biopsy with **acid-fast staining**.



Culture, PCR, and sequencing may be helpful in identifying the mycobacterial species. Culture can take a long time as these organisms are fastidious growers.

Treatment is difficult and most of the time these patients are euthanized. There is also a potential for zoonosis as *Mycobacterium marinum* for instance is known to cause skin lesions in people. The treatment consists of the use of a combination of antimycobacterial antibiotics chronically so owners have to be able to medicate the animal for a long period of time. There are not a lot of report of treatment of reptile mycobacteriosis actually.

Avian mycobacteriosis

Avian mycobacteriosis is also not uncommon. It primarily is a **disease of the gastrointestinal and hepatic system**. The species are different than seen in reptiles. The main species in wild and zoo birds tend to be ***Mycobacterium avium avium***, which is difficult to distinguish from *Mycobacterium intracellulare* (that's why they are grouped under the term of *M. avium-intracellulare* complex). The main species in companion birds is ***Mycobacterium genavense***. As for reptiles, these are saprophytic organisms and are birds are typically infected from environmental sources rather than from other infected birds. Immunocompromised individuals are at more risks. In zoos, filtering species such as ducks are also more at risks in a contaminated environment. Overall, all bird species are susceptible to avian mycobacteriosis.

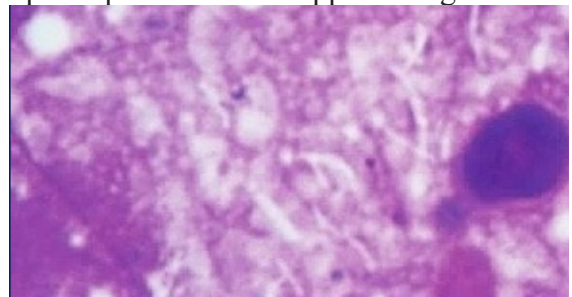
Birds are also susceptible to tuberculous mycobacteria such as *Mycobacterium tuberculosis* and *Mycobacterium bovis* that they typically get from infected humans (reverse zoonosis). They will develop granulomas, typically on the head where they are kissed or petted by their infected owners.

The clinical signs are relatively non-specific, but this is often a **chronic presentation with significant weight loss**. The GI, liver and sometimes respiratory system can be affected.

On hematology, there is typically a **very high inflammatory response with a high heterophilic leucocytosis and monocytosis**. Those numbers are some of the highest I have seen, along with leukemia. It can also be normal especially for localized disease, so a lack of leukocytosis does not rule it out.

On imaging, granulomas can be seen in some organs, typically GI and liver, but could also be present in other organs. Imaging is generally not very sensitive unless the disease is advanced.

The best way to diagnose this disease is to sample the lesions or the liver or spleen. Cytology or histopathology with acid-fast staining will show these pathogens. On cytology with standard stains, these bacteria do not pick up the stain and appear as “ghost” bacteria.



A **PCR on tissue** (liver or spleen being better) is also good. Culture takes a lot of time so it is rarely done, unless there are acid-fast bacteria with a negative sample for *M. avium* and *M. genavense*. **PCR on cloacal swabs or feces is minimally useful** as these organisms are normally found in the environment and could just be pass-through organisms.

For advanced disease, euthanasia should be recommended as it is difficult to treat. These pathogens have **zoonotic potential, but a bird-to-human transmission has not actually been documented to date.** So if the disease is mild to moderate and the household does not include immunocompromised people, treatment is definitely an option. Treatment requires the use of a combination of antimycobacterial antibiotics that could include isoniazid, clarithromycin, ethambutol, rifampin, quinolones and tetracyclines. **A triple therapy including clarithromycine-ethambutol-rifampin** is frequently used as a first line. Birds have to be treated for months to years and the disease tends to come back at some point, sometimes years later.

Adenovirus in bearded dragons

The official species name of this virus is *Barthadenovirus draconis*, but it is most often called bearded dragon adenovirus 1, agamid atadenovirus 1, or lizard atadenovirus B in the literature. The virus and disease are specific to bearded dragons (*Pogona vitticeps* at least).

There seems to be a **high prevalence and many animals are subclinical for it.** This means you should test bearded dragons in quarantine for it. The disease is **mainly seen in juvenile** bearded dragons and is characterized by gastrointestinal and neurological signs. Hepatitis is also frequently seen with intranuclear inclusion bodies on histopathology.



The diagnosis is made by an orocloacal PCR. There is no specific treatment so far.

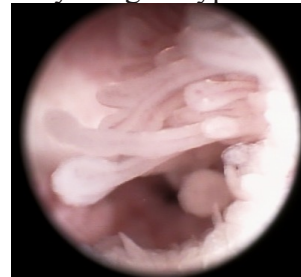
Pacheco's disease

This virus actually primarily targets the liver and is a very important disease of parrots.

The taxonomic name of this virus is *Iltovirus psittacid-alpha 1*, but it is most commonly known as **psittacid herpesvirus 1**. It is an alphaherpesvirus and there are 4 main genotypes and several serotypes. There are two disease forms that you should remember.

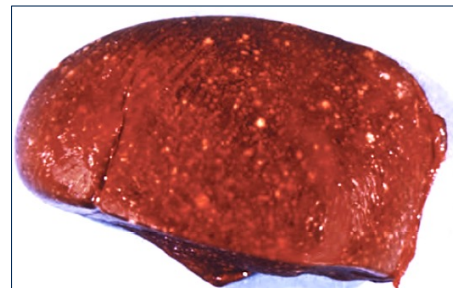
The first is an **acute liver necrosis that is highly contagious and leads to high mortality in aviaries**, this is Pacheco's disease. The disease is **more common South American parrots with Amazon parrots** being the poster chick (avian equivalent of the poster child) for the disease.

The second form is seen in subclinical carriers where the virus in its latent form leads to cloacal papillomas (most South-American species) or esophagus (green-winged macaws). This is **internal papillomatosis**. These birds can also develop carcinomas in the biliary or pancreatic duct. This disease manifestation seems to be mainly caused by the genotype 3 of the virus.



In any case, **any bird that does not die from Pacheco's disease is permanently infected** and the virus goes into a latency form. **Under stress, birds may start shedding the virus again** leading to disease in other birds. Conures (especially Patagonian conures) have been implicated as frequent subclinical carriers of this virus and it is recommended to never mix conures with other species in a mixed species aviary.

Since it has a high mortality rate and is a fulminant disease, **any die-off in an aviary of parrots should raise a strong suspicion for Pacheco's disease.** Diagnostics are typically not feasible before the bird's death as it is too rapid. On necropsy, hepatic necrosis with intranuclear inclusion bodies is seen.



A PCR from an orocloacal swab can be performed for ante-mortem diagnosis.

Outbreaks can be managed with acyclovir, which seems to really decrease mortality rate in an aviary. This has to be given orally to each individual bird for several days. It should be started prior to having confirmation of the disease.

Duck viral enteritis

This is caused by *Mardivirus anatis*, or most commonly known as **anatid herpesvirus 1 or duck plague**. This is an important disease of wild and commercial ducks. It is highly contagious and leads to **high mortality rate and potential large die-off in the wild**. For die-off in the wild, **other differentials include pasteurellosis and botulism**, but in the case of duck plague only ducks, geese, and swans are affected and coots and other shorebirds are spared. This is more common along the Atlantic flyway.

The main target is actually the **gastrointestinal tract** (hence the name), but the liver is also affected and inclusion bodies will be seen on histopathology. The disease is peracute with intestinal and hepatic necrosis and DIC. **“Cold sore”** under the tongue can be seen in some cases and is a good indication in the wild that a herpesvirus outbreak is happening before laboratory confirmation.



As for all herpesviruses, survivors are persistently infected and the virus enters a latent form. The main wild reservoir seems to be the **Northern Pintail** but other ducks have been implicated in the disease's epidemiology such as **mallards**.



Columbid herpesvirus

The name of the virus is actually *Mardivirus columbidalpha 1*, but it is most commonly known as **columbid herpesvirus 1**. As we initially did not know that it was the main cause of herpesvirus hepatitis in raptors, there are lots of other names in the older literature for this virus including falcon herpesvirus, owl herpesvirus, and accipitrid herpesvirus. If you go way back in the literature, it was also called Smadel's disease in pigeons.

The infection is **very common in pigeons, but few show clinical signs, mainly youngsters**. Raptors get infected by preying upon these poor pigeons and also other columbids. The infection is essentially a **spill-over infection** of a virus adapted to pigeons and causing minimal clinical signs in the host species (as it is the case for many herpesviral diseases), but causing fulminant necrotic hepatitis with inclusion bodies and other signs such as oral plaques and high mortality in non-adapted species.



Diagnosis is difficult prior to death in most cases, but could be performed using PCR. Treatment with acyclovir can be attempted.

Systemic isosporosis in passerines

This disease was previously known as **atoxoplasmosis** and you may find this term quite a bit in the older literature and even in the contemporary literature. What we know now is that this disease is actually caused by a number of **different Isospora species** in passerine birds that seems somewhat species specific. The one in canaries is called *Isospora serini*. The transmission is fecal-oral and the life cycle is direct meaning that infection can escalate quickly in captivity.

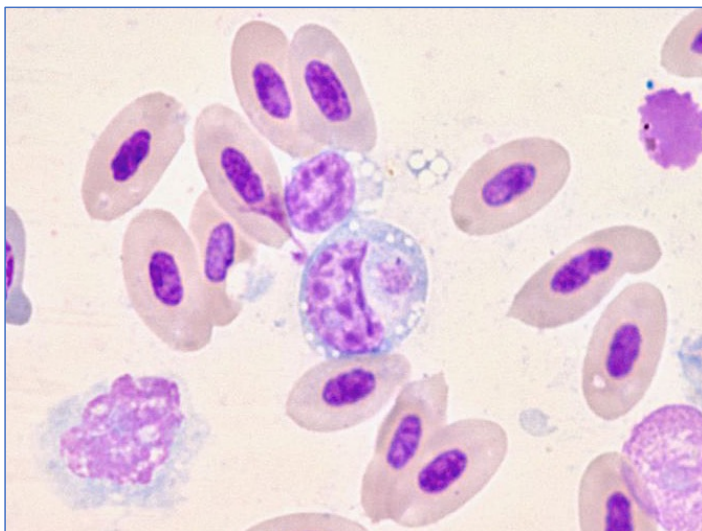
Signs are usually non-specific but the disease **targets the spleen, the liver, other organs, and mononuclear**

inflammatory cells. This leads to inclusions visible in the blood in some cases as well as splenomegaly and hepatomegaly. In passerines, the skin is very thin and a bit transparent so you may actually see the liver going passed the keel and the infected spleen as a black dot. The disease layman name is **black spot disease** in canaries. During physical examination, you can also **transilluminate the abdomen** of small birds to see the liver and spleen better.



Clinical signs are most common in chicks and juvenile birds and subclinical in adults. It is also an important disease to know about in some bird conservation program such as in the **Bali Mynah**.

On blood smear or buffy coat smear, **inclusions may be detected in mononuclear cells.**



A **PCR** is also available depending on the species. Fecal flotation will also show *Isospora* oocysts, but there are also some *Isospora* that stays in the GI. On necropsy, impression smears of organs will show inclusions as well.

The treatment is actually difficult and the disease may not respond well to anticoccidial treatment. Drugs that can be tried include **sulfachlorpyrazine, toltrazuril, or ponazuril.** **Cleaning very well the environment, in particular the substrate, is also very important to get rid of oocysts.**

Encephalitozoon pogonae

Encephalitozoon pogonae is a **microsporidia**, closely related to fungi, that causes a **multisystemic granulomatous disease** in bearded dragons. Lesions can be anywhere and frequently include the liver and may also manifest as **intracoelomic masses/granulomas.**

Epidemiologic studies have shown that these bearded dragons are also frequently co-infected with adenovirus, so it is possible that it predisposes them to the infection.

Signs, as usual with reptiles, are non-specific and mainly characterized by anorexia and lethargy. Antemortem diagnosis is tricky and we actually don't have a lot of information on this but PCR testing is available. Oro-cloacal swabs or tissues can be submitted.

The treatment is poorly reported, but people have tried using **fenbendazole.**

NON- INFECTIOUS HEPATOPATHIES

Many non-infectious diseases have been reported in reptiles. Hepatic lipidosis, hepatic fibrosis, amyloidosis, visceral gout, neoplasia (primary or metastasis) are seen in both taxa. We'll spend a bit more time on hepatic lipidosis as it is common in both reptiles and birds.

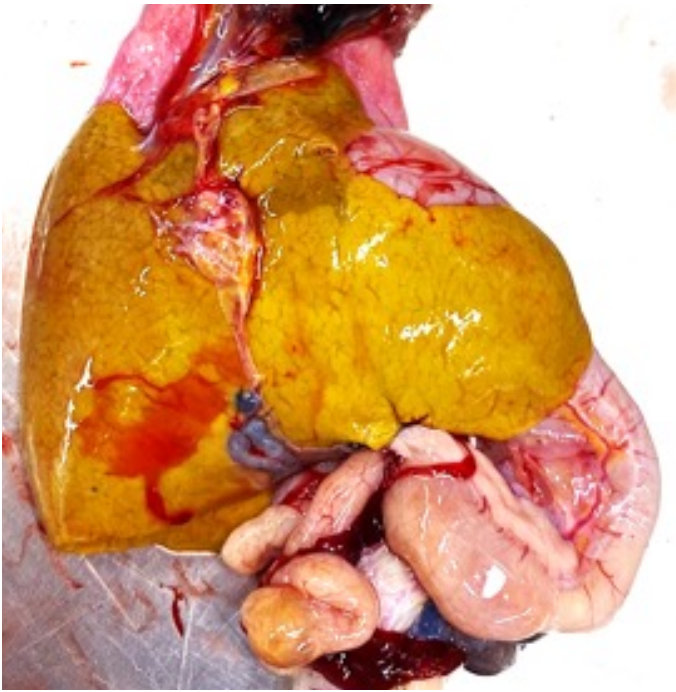
Lymphoma is a common neoplasia in reptiles and birds. Primary cancer such as hepatic adenoma and adenocarcinoma have been seen in birds. Cholangiocarcinoma and pancreatic duct carcinoma are seen in South-American parrots likely as consequence of psittacine herpesvirus infection. In bearded dragons, metastasis of gastric neuroendocrine tumors to the liver is common with this cancer.

In birds, iron storage disease is common in specific species and warrants more discussion. Hepatic toxins can also be encountered. I would say that the most common are mycotoxins, typically related to seeds and nuts.

Hepatic lipidosis in reptiles

Hepatic lipidosis (or rather hepatic fat accumulation)

is exceedingly common in a wide range of reptiles, especially in **bearded dragons**, where the prevalence can be as high as 30%. It is characterized by the **progressive accumulation of triglycerides in hepatocytes resulting in an enlarged friable yellow liver**. It is currently debated at which stage of fat accumulation it becomes pathologic (lipidosis). It seems that many reptile species, and especially bearded dragons, have an incredible ability to store fat in the liver with minimal consequences and for various physiological reasons, one of them being vitellogenesis in females. So it is not clear when hepatic fat accumulation becomes a disease in many of these species. Nevertheless, in some individuals, the accumulation is just too much and may lead to **hepatic dysfunction, impaired metabolism, and hepatic fibrosis**.



The pathogenesis is fundamentally different from what you know in cats and cows in that it is not caused by acute anorexia in an overweight animal. The causes and pathogenesis are not well defined and probably vary by species. **The disease is mainly seen in adults with females slightly predisposed. Risk factors seem to include dietary (high fat, high carbohydrates, high feeding frequency), environmental (low temperature), and lifestyle (sedentarity, lack of brumation and reproduction) factors.**

It has been difficult to attribute particular clinical signs to hepatic lipidosis as it is concurrent with other diseases and lesions, being so common. It seems that anorexia and lethargy in absence of other causes would make it

more likely to be linked to hepatic lipidosis.

On blood work, there is almost no changes seen in bearded dragons and other lizards but hypertriglyceridemia has been reported in some chelonians. In bearded dragons, low **BHBA** has consistently been shown to be present with severe hepatic lipidosis. Other hepatic parameters are typically unaltered, but changes could be seen in end-stage disease with hepatic dysfunction.

Ultrasound is not particularly reliable, mainly because it is hard to conclusively characterize hyperechogenicity in reptiles or just quantify the level of fat, but ultrasound could be suggestive, especially if the liver is enlarged.

CT-scan is the best non-invasive diagnostic test as the density of the liver is linearly related to its fat content. A liver with low attenuation (<20HU) is almost pathognomonic for hepatic lipidosis.



However, **only hepatic biopsy can help with staging and assessing hepatic fibrosis** and ruling out other

hepatopathies. Hepatic biopsies can be obtained either by laparotomy or endoscopic biopsy.

In terms of treatment, we do not have good confirmatory evidence of what is working. The current recommendations include:

- **Supportive care:** fluid therapy and assisted feeding are important in debilitated and anorexic animals. Most commercial critical care formula should be avoided for this disease as they are often extremely high in carbohydrates and reptiles become severely hyperglycemic on those, which does not help with fatty liver disease. It is better to just use a slurry of greens and a few insects supplemented with calcium carbonate. An esophagostomy tube can also be placed when chronic enteral feeding is necessary.

- **Dietary changes:** reducing fat dietary items such as larval insects and other high fat insects. Increasing proportion of greens for omnivorous species. Decreasing carbohydrates such as fruits and decreasing feeding frequency (not every day).

- **Environmental changes:** increasing temperature. Bearded dragons need basking spots close to 105 F. High temperature help with metabolizing fat. Increasing foraging opportunities and exercise.

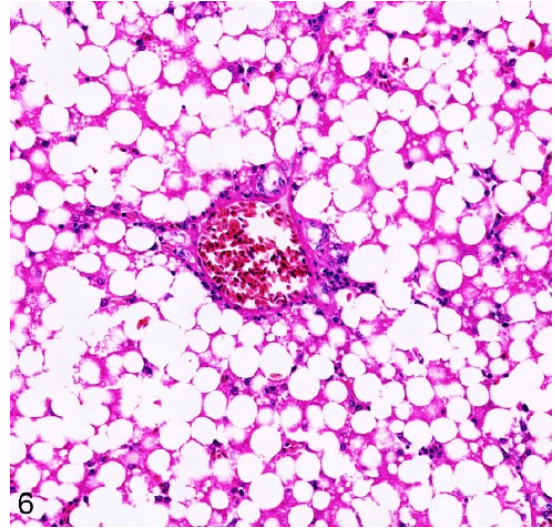
- **Drugs:** lipotropic factors can be given such as carnitine which helps with fatty acid oxidation. **S-adenosyl-methionine** (same) is an hepatoprotectant and helps scavenge free radicals generated during lipid fat peroxidation, which is common in fatty liver. **Gemfibrozil** is a fibrate drug that directly promotes fatty acid oxidation in the liver mitochondria and we are using it more and more, but still not sure whether it truly helps.

Hepatic lipidosis in birds

Hepatic lipidosis is also very common in birds and bears a lot of similarities with its reptilian counterpart. Again, it is not typically caused by anorexia episodes in obese animals and seems to be a dietary issue. The disease appears to have a different pathogenesis depending on species, especially when comparing chickens to other birds.

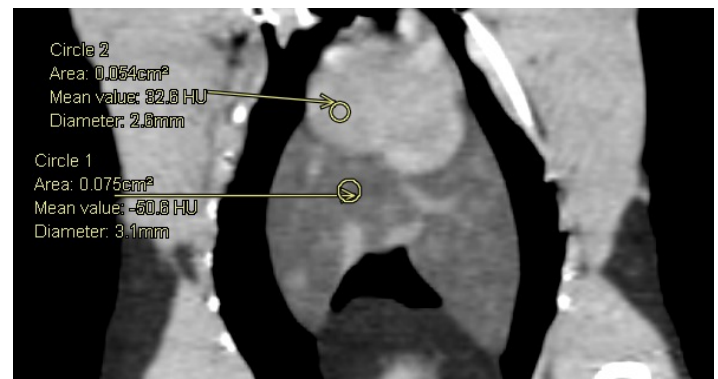
In psittacine birds, it is relatively common with an overall prevalence of 6%, but more in susceptible species such as Quaker parrots and Amazon parrots. It is common in chickens too and is strongly associated with female vitellogenesis and egg laying as estrogens stimulate fat production in the liver. This sex predisposition does not seem to be true in parrots. The disease is also seen in a variety of other birds kept under human care such as raptors. The liver is naturally

steatotic in chicks so they should not be misdiagnosed with hepatic lipidosis.



Risk factors in parrots are poorly defined but it is more common with a **high fat diet and poor overall activity**. Likewise, in egg-laying chickens, fatty liver is associated with high energy diet and exercise restriction and is common in caged laying hens and obese backyard chickens. Actually, a **diet high in carbohydrates is more predisposing than a high-fat diet** (like in most species and humans actually). There is a particular syndrome in chickens called the **fatty-liver hemorrhagic syndrome**. It seems that a large steatotic liver is very friable and during egg laying, it can rupture and lead to fatal hemorrhage.

Like for reptiles, biochemistry is mostly unremarkable until significant hepatic dysfunction is present. It is unclear whether hyperlipidemia is seen concurrently, but it seems to be the case in chickens at least. Diagnostic imaging is identical to reptiles with **CT being overall superior** to make the diagnosis (see below a coronal view in a parrot showing a low attenuating liver compared to the heart). However, a **liver biopsy is necessary** for a confirmatory diagnosis or to characterize non-lipid changes such as inflammation and fibrosis.



Treatment is also similar to reptiles and based on correcting the diet, increasing activity, and using hepatoprotectant drugs and lipotropic factors. Supportive care should be administered as required.

Iron storage disease

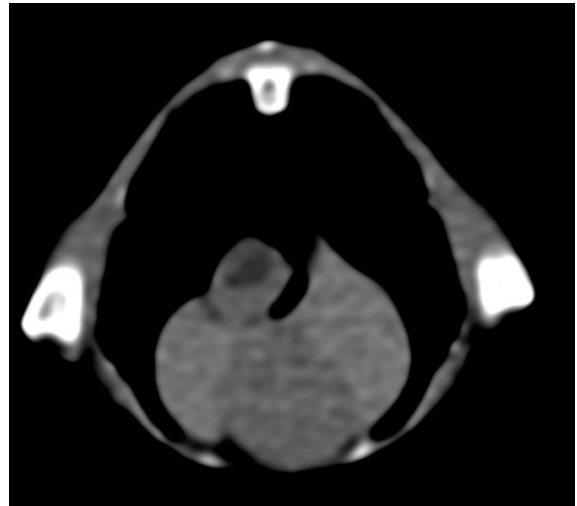
Iron storage disease is caused by the excessive accumulation of iron in the liver in iron-sensitive species. These species include mainly **toucans, mynahs, starlings** (and other sturnidae), birds of paradise, lorikeets, hornbills, and many frugivorous species. Supposedly, these species have evolved in a low-iron environment, but there are other species-specific factors involved. For instance, it has been shown in the starling that they are very efficient at absorbing iron in their GI.

Hepatic lesions are called hemosiderosis when there is just an accumulation of iron without pathologic changes and hemochromatosis when hepatocytes start looking funky.

Clinical signs are those associated with liver disease. In mynahs, hepatic insufficiency is common and you may see ascites and other signs of hepatic insufficiency. Iron accumulation also leads to **cardiomyopathy**, mainly reported in mynahs, and **diabetes mellitus** as it is toxic to the pancreas as well.

The diagnosis in birds is tricky and harder than in susceptible mammals (such as lemurs, rhinos and so on). **Iron metabolism parameter in the blood are not sensitive at all in birds.** Those include TIBC (total iron binding capacity), serum iron, and transferrin. It can only be performed using liver biopsy and special staining (Prussian blue stain). MRI may be able to quantify hepatic iron, but is not a practical solution in most circumstances and is very expensive. Sometimes with

severe lesions, you can see an increase hepatic attenuation on CT-scan as iron has a high atomic number compared to other atoms found in hepatic molecules. See an example below in a roller housed in a zoo and fed an inappropriate diet. Heart is central and of lower attenuation.



In terms of treatment, the goal is to decrease the amount of iron. **Periodic phlebotomy** of 1% of the body weight weekly or every other week can be performed and is very effective, but may not be practical in fractious species. **Iron chelating agents such as oral deferiprone or injectable deferoxamine** can also be used, but are expensive. Putting birds and iron deficient diet may also be implemented. Soaking food in decaffeinated black Ceylan tea is supposedly decreasing iron content.

Prevention is key in this disease as it will happen in these species otherwise. **Feeding a low iron diet is essential.** **Citrus fruits** also promote iron absorption and should be fed in limited quantity especially when fed at the same time as other food items.



Male superb bird of paradise trying to impress a female just passing by.

An unimpressed female superb bird of paradise

The Ten Most Important Facts

1. The liver in birds and reptiles has two lobes with the right generally larger and harboring the gallbladder. Most reptiles have a gallbladder, but, within birds commonly seen in practice, parrots and pigeons don't. Cockatoos, cockatiels, and raptors have a gallbladder.
2. Clinical signs of hepatic disease in sauropsids are typically non-specific and subtle, including anorexia, lethargy, weight loss, polyuria, and biliverdinuria, with icterus being absent.
3. Hepatic biochemistry evaluation requires a panel composed of several enzymes such as AST, LDH, GGT, GLDH, and CK as well as bile acids and metabolites. AST and LDH are not liver specific but are sensitive while GLDH is highly liver specific, but is not sensitive. Bile acid is a functional test and is both sensitive and specific.
4. Radiographs are mainly useful in birds by evaluating the cardiohepatic silhouette while it is difficult to evaluate in reptiles. Ultrasound is a great diagnostic test for liver diseases and also allow fine-needle aspiration and cytology of lesions. CT-scan is also very valuable, especially with contrast, and allows to assess hepatic fat content.
5. Mycobacteriosis is an important infectious hepatopathy in both birds and reptiles, often causing chronic granulomatous disease and requiring PCR or acid-fast staining for diagnosis, with long and often unrewarding treatment courses. *Mycobacterium marinum* and *chelonae* are the most common species in aquatic turtles while *Mycobacterium avium avium* and *genavense* are the most common in birds.
6. Adenovirus in bearded dragons is a common infection mainly leading to clinical signs in juvenile animals with adults having a high prevalence of subclinical carriage state. Clinical signs include gastrointestinal and neurologic signs. Diagnosis is made by PCR and histopathology and no treatment is currently available.
7. Pacheco's disease, caused by an alphaherpesvirus, is a major cause of acute hepatic necrosis and mortality in parrots, with latency and stress-induced recrudescence, and PCR and histopathology are essential for diagnosis during outbreaks. Acyclovir can be used to mitigate mortality in an aviary.
8. Systemic isosporosis is encountered in a variety of passerine species, most notably the canary, and is caused by various *Isospora* spp. Treatment is difficult and consists of anticoccidial medications such as ponazuril.
9. Hepatic lipidosis is highly prevalent in reptiles and birds, often associated with poor diet, obesity, or reproductive status, and is best diagnosed through imaging (especially CT) and confirmed with liver biopsy. Some species kept as companion animals are particularly prone such as bearded dragons, Amazon parrots, and Quaker parrots.
10. Iron storage disease is a major concern in frugivorous bird species like toucans and mynahs, requiring histopathologic diagnosis and managed through dietary modification, iron chelation, or periodic phlebotomy.